Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_40

1. Summary of Key Deviations

The analysis of Run_40 reveals a critical and highly localized fault. The system exhibits a clear pattern of flow being redistributed due to a severe downstream blockage affecting a single component.

- **Flow_Engine1 (Critical):** This is the primary and most severe anomaly. The signal for engine 1 shows a **major negative drift** culminating in a complete **signal dropout (-99.7%)** in state s5. Flow is almost entirely cut off to this specific engine nozzle.
- **Flow_Bypass (Moderate):** A large positive **offset** (up to **+33.2%**) is observed. This is a direct physical consequence of the blockage at engine 1; with its primary path obstructed, flow is forced back through the bypass loop.
- **Flow_Pump (Moderate):** The pump shows a moderate positive **offset** (up to **+17.5%**). This is likely the control system's response, increasing total flow to try and compensate for the blockage and maintain system pressure and flow to the other engines.
- **Flow_Engine2, 3, 4 (Nominal):** Crucially, the other three engine flow sensors show **no significant deviation**. They exhibit minor transients or offsets but remain fundamentally healthy. This proves the fault is isolated to the branch feeding engine 1.
- **Flow_PRV & Pressures (Nominal):** There are no significant deviations in system pressures or PRV flow, indicating the system is successfully managing the event without a dangerous over-pressurization.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Injector Nozzle 1**	**Severe Blockage / Clogging**	This is the most likely root cause. A near-total blockage of the first injector nozzle perfectly explains why **Flow_Engine1** drops to zero while **Flow_Engine2, 3, and 4** remain healthy. This single-point failure forces all the un-delivered fuel to be redistributed, causing the large increase in **Flow_Bypass** and prompting the **Flow_Pump** to work harder.

2	**Flow_Engine1 Sensor**	**Sensor Dropout / Failure**	It's possible the sensor for engine 1 failed and is incorrectly reporting zero flow. However, this is less likely because the **Flow_Bypass** and **Flow_Pump** readings are both physically consistent with a real blockage. If the sensor simply failed, there would be no reason for the bypass flow to increase so dramatically.
3	**Fuel Metering Unit (FMU)**	**Internal Blockage (Channel to Engine 1)**	The FMU has separate channels or ports that feed each engine line. A blockage or failure specific to the channel feeding engine 1 would produce identical symptoms to a clogged nozzle. This is a very strong alternative to a clogged nozzle.
4	**Piping to Engine 1**	**Line Crimp / Blockage**	The physical fuel line leading to engine 1 could be crimped or blocked. This is physically equivalent to a clogged nozzle or a blocked FMU channel and is another plausible location for the same root fault.
5	**System Contamination**	**Debris causing blockage**	A piece of debris could have become lodged in the fuel line, FMU port, or the nozzle for engine 1. This is the underlying cause of the blockage itself, but the *location* of the blockage is the key diagnostic finding.

3. Most Probable Hypothesis

The most probable root cause is a **Severe Blockage of the Fuel Line or Injector Nozzle for Engine 1.**

Reasoning: The evidence points definitively to a severe or total blockage isolated to the fuel path of Engine 1. The critical dropout of the `Flow_Engine1` signal, combined with the complete health of the other three engine signals, pinpoints the failure location to somewhere between the FMU's split and the tip of injector nozzle 1. The large, compensatory increases in `Flow_Bypass` and `Flow_Pump` are the physically necessary reactions to this blockage and serve as strong secondary evidence that this is a real physical fault, not just a sensor failure.